COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	02/09/2026	**DATE SUBMITTED:**	02/09/2026
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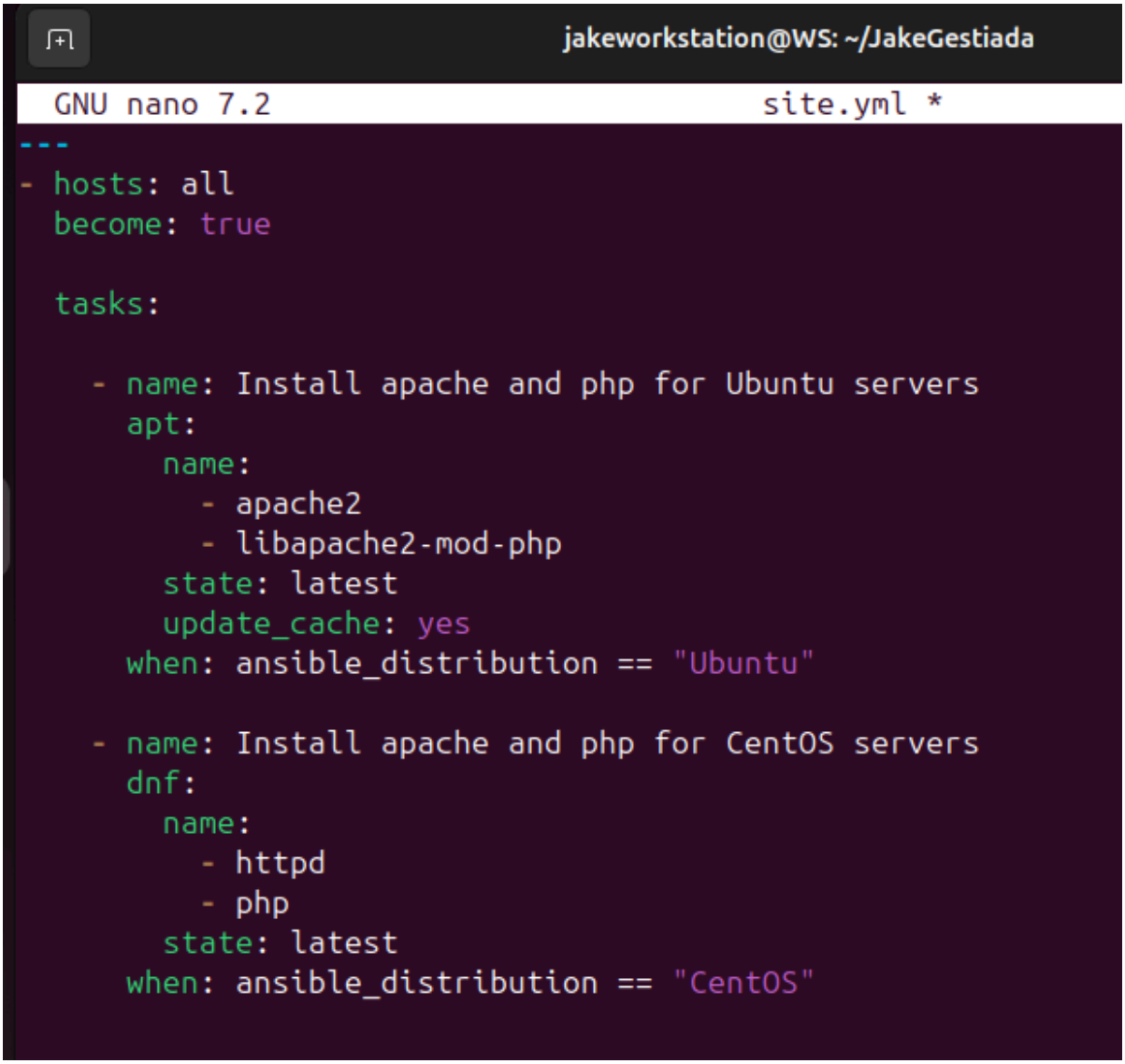
ACTIVITY NO. 6.1 **Targeting Specific Nodes**

1. PROCEDURE OUTPUT

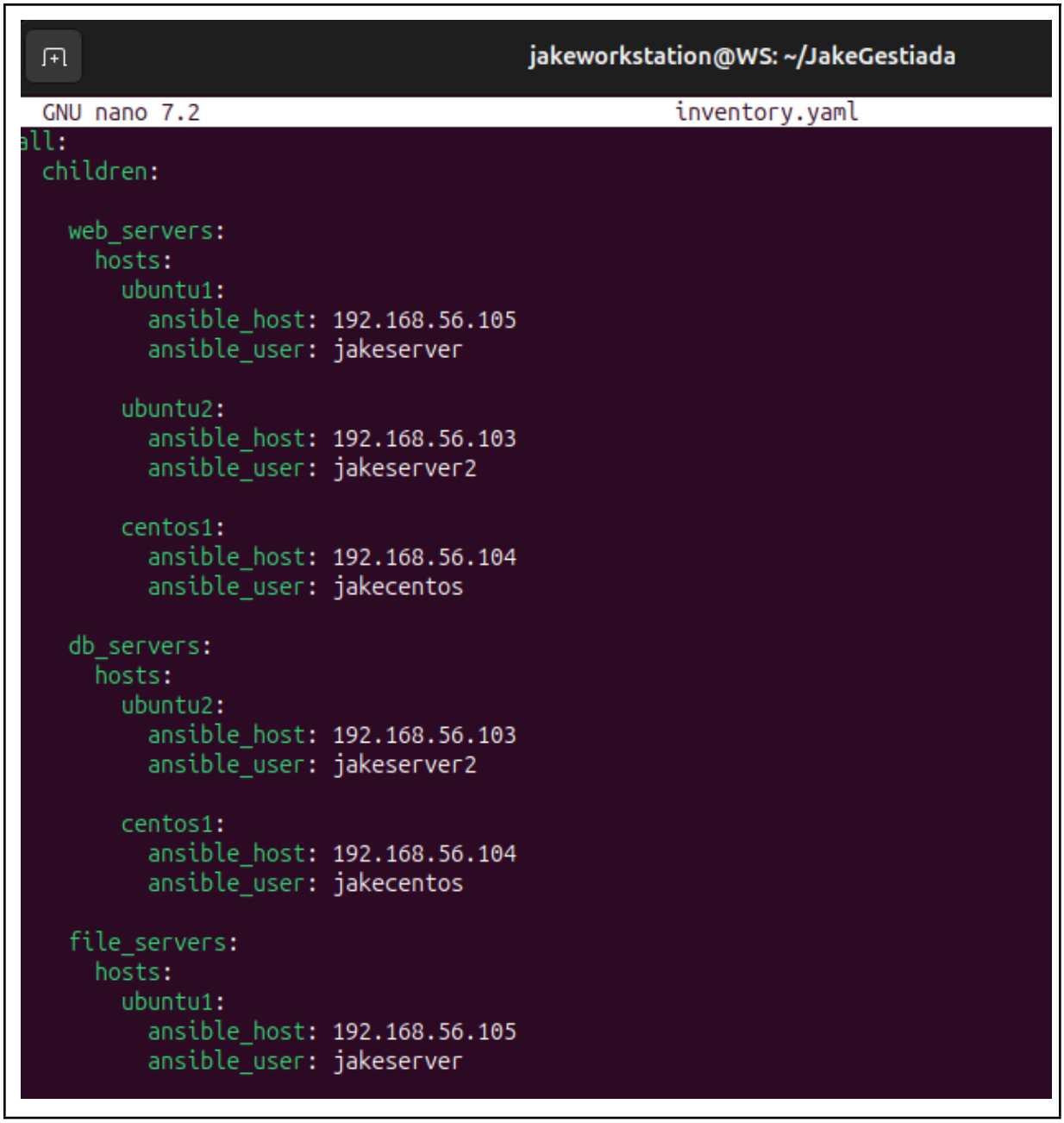
What procedures did you perform?

Task 1: Targeting Specific Nodes

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.



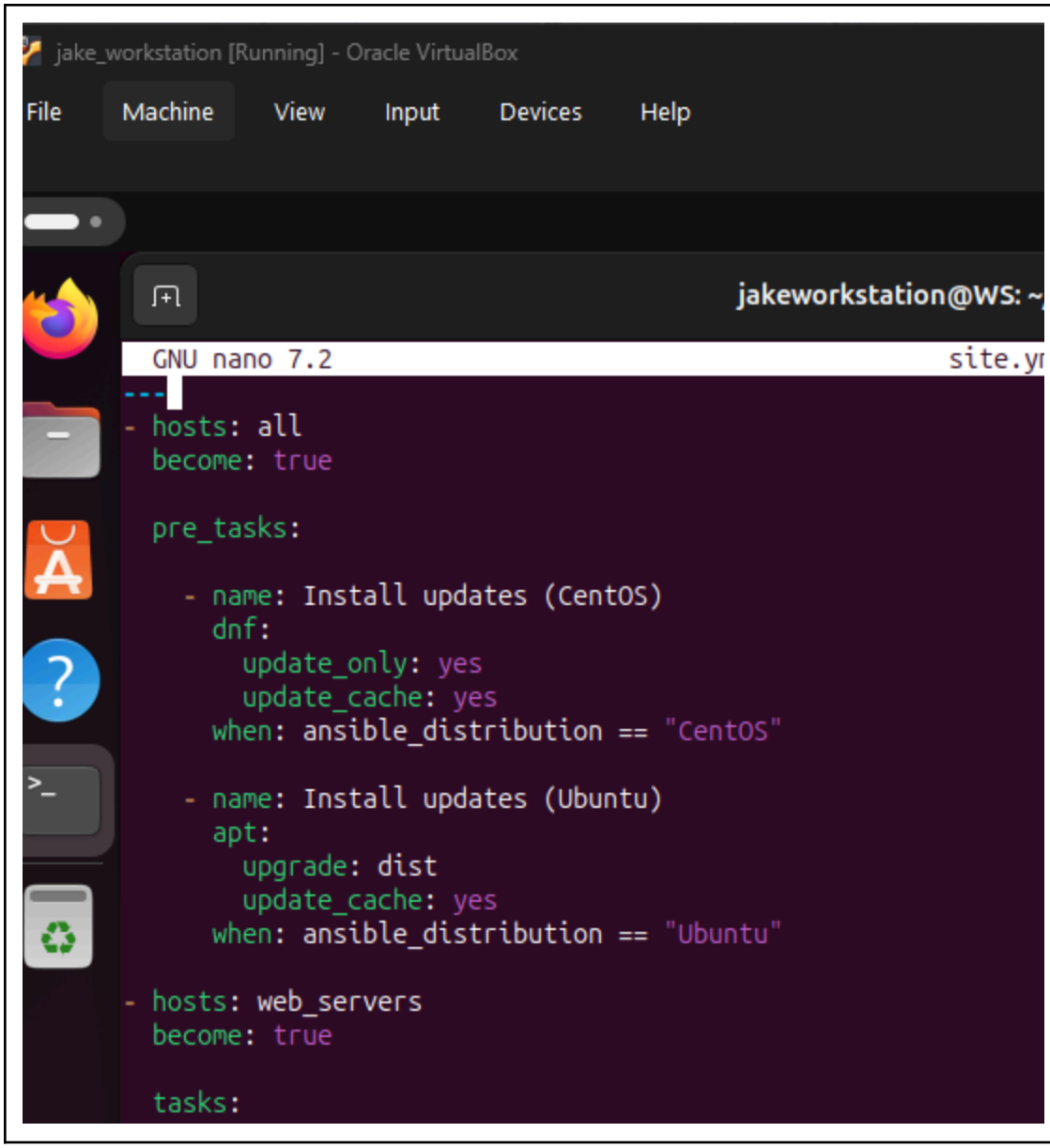
2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:
Make sure to save the file and exit.
Right now, we have created groups in our inventory file and put each server in its own group. In other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server


jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2
inventory.yaml
all:
children:
web_servers:
hosts:
ubuntu1:
ansible_host: 192.168.56.105
ansible_user: jakeserver
ubuntu2:
ansible_host: 192.168.56.103
ansible_user: jakeserver2
centos1:
ansible_host: 192.168.56.104
ansible_user: jakecentos
db_servers:
hosts:
ubuntu2:
ansible_host: 192.168.56.103
ansible_user: jakeserver2
centos1:
ansible_host: 192.168.56.104
ansible_user: jakecentos
file_servers:
hosts:
ubuntu1:
ansible_host: 192.168.56.105
ansible_user: jakeserver

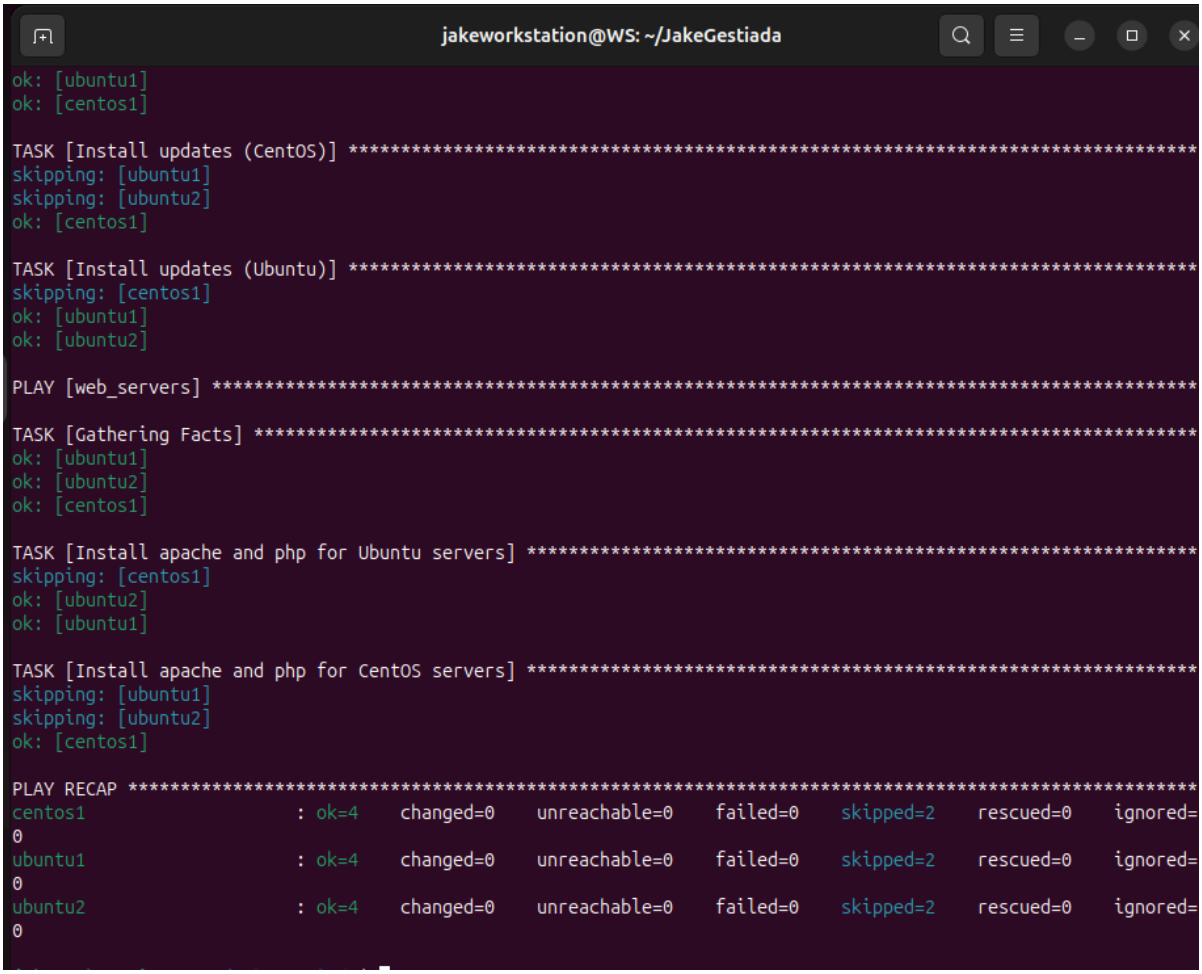
```
jakeworkstation@WS:~/JakeGestiada$ ansible-inventory -i inventory.yaml --graph
@all:
  |--@ungrouped:
  |--@web_servers:
  |  |--ubuntu1
  |  |--ubuntu2
  |  |--centos1
  |--@db_servers:
  |  |--ubuntu2
  |  |--centos1
  |--@file_servers:
  |  |--ubuntu1
jakeworkstation@WS:~/JakeGestiada$
```

```
jakeworkstation@WS:~/JakeGestiada$ ansible web_servers --list-hosts -i inventory.yaml
  hosts (3):
    ubuntu1
    ubuntu2
    centos1
jakeworkstation@WS:~/JakeGestiada$ ansible db_servers --list-hosts -i inventory.yaml
  hosts (2):
    ubuntu2
    centos1
jakeworkstation@WS:~/JakeGestiada$ ansible file_servers --list-hosts -i inventory.yaml
  hosts (1):
    ubuntu1
```

3. Edit the site.yml by following the image below:
Make sure to save the file and exit.
The pre-tasks command tells the ansible to run it before any other thing. In the pre-tasks, CentOS will install updates while Ubuntu will upgrade its distribution package. This will run before running the second play, which is targeted at web_servers. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.
Run the site.yml file and describe th


jake_workstation [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
jakeworkstation@WS: ~
GNU nano 7.2
site.yr

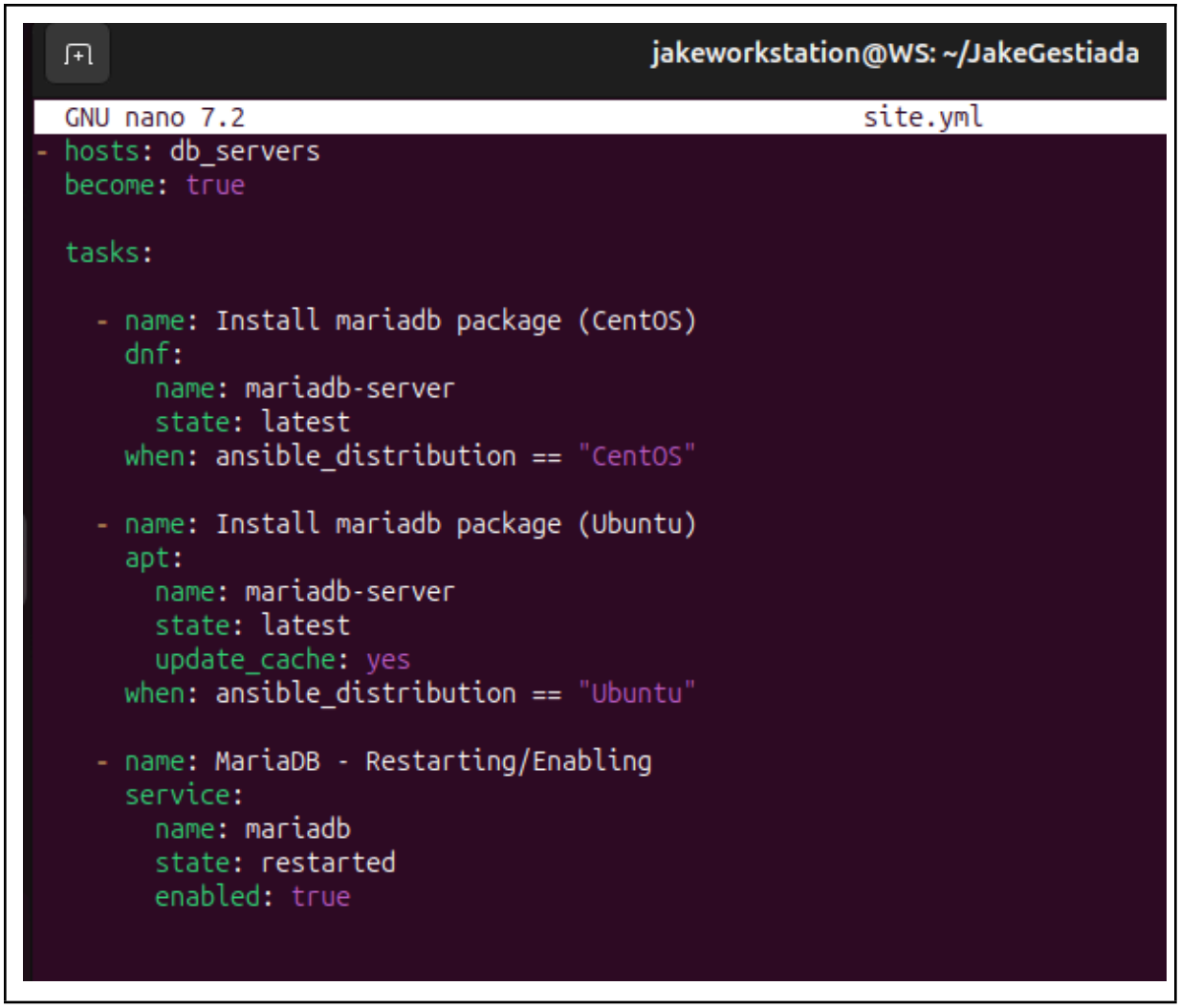
- hosts: all
become: true
pre_tasks:
- name: Install updates (CentOS)
dnf:
update_only: yes
update_cache: yes
when: ansible_distribution == "CentOS"
- name: Install updates (Ubuntu)
apt:
upgrade: dist
update_cache: yes
when: ansible_distribution == "Ubuntu"
- hosts: web_servers
become: true
tasks:

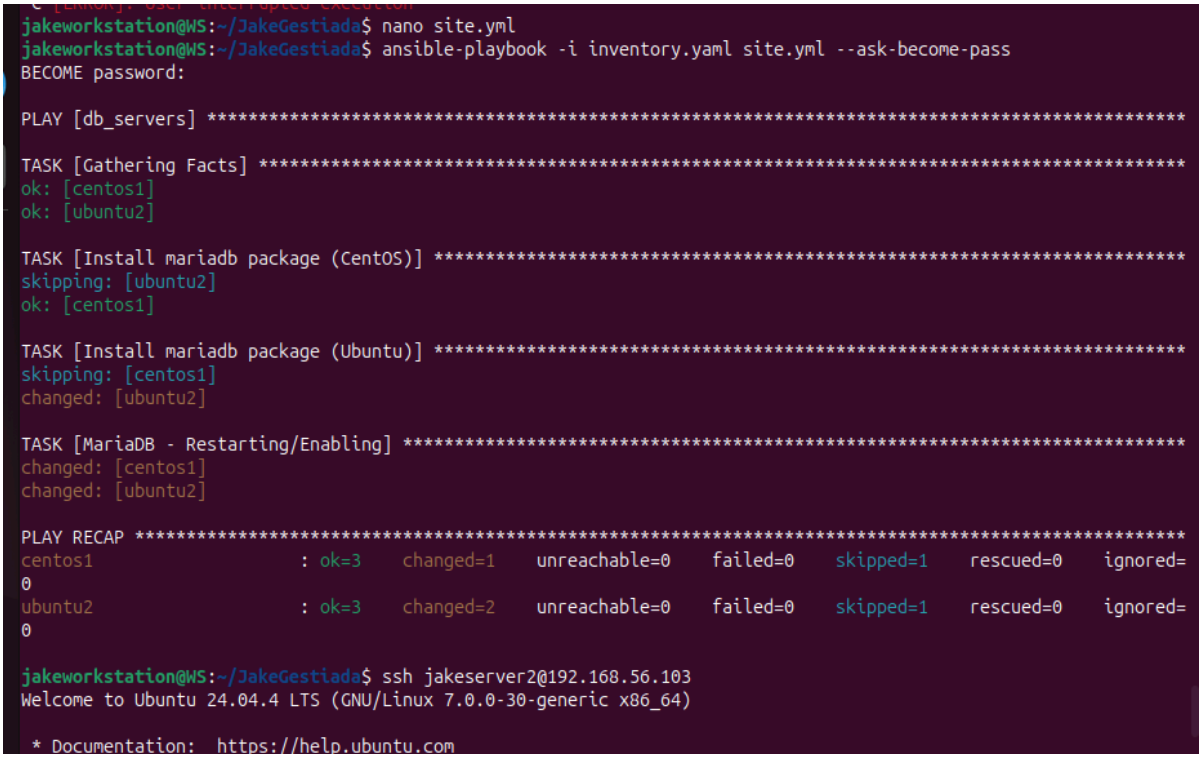


Describe the result:

- This is done by the playbook to update the servers according to their respective operating systems. It installed Apache and PHP on the web servers. While Ubuntu relied on apt for its updates, CentOS used dnf. Thanks to the when conditions, the tasks were executed only on the right servers that matched the condition.

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers. This time we target the db_servers by adding it on the current site.yml. Below is an example: (Note add this at the end of the playbooks from task 1.3. Make sure to save the file and exit. Run the site.yml file and describe the result.

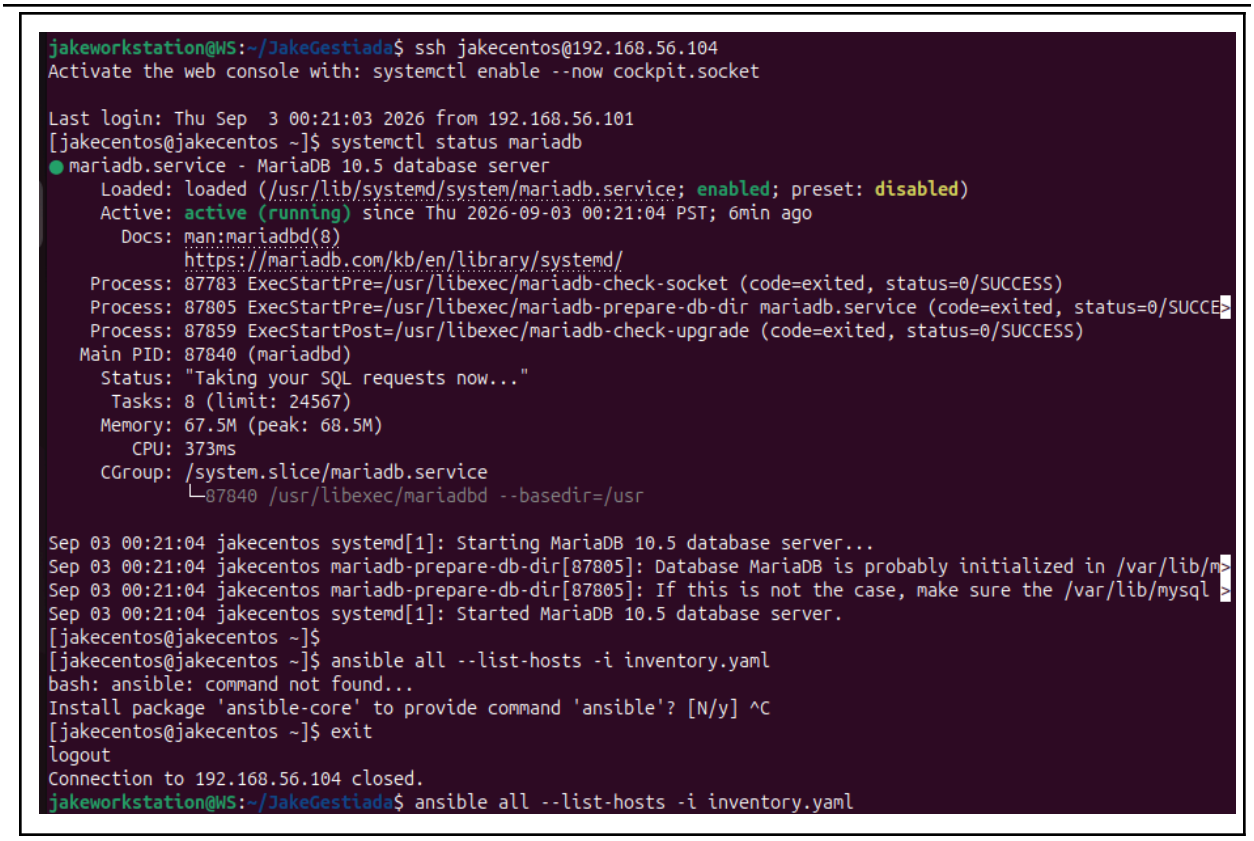

jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2 site.yml
- hosts: db_servers
become: true
tasks:
- name: Install mariadb package (CentOS)
dnf:
name: mariadb-server
state: latest
when: ansible_distribution == "CentOS"
- name: Install mariadb package (Ubuntu)
apt:
name: mariadb-server
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: MariaDB - Restarting/Enabling
service:
name: mariadb
state: restarted
enabled: true

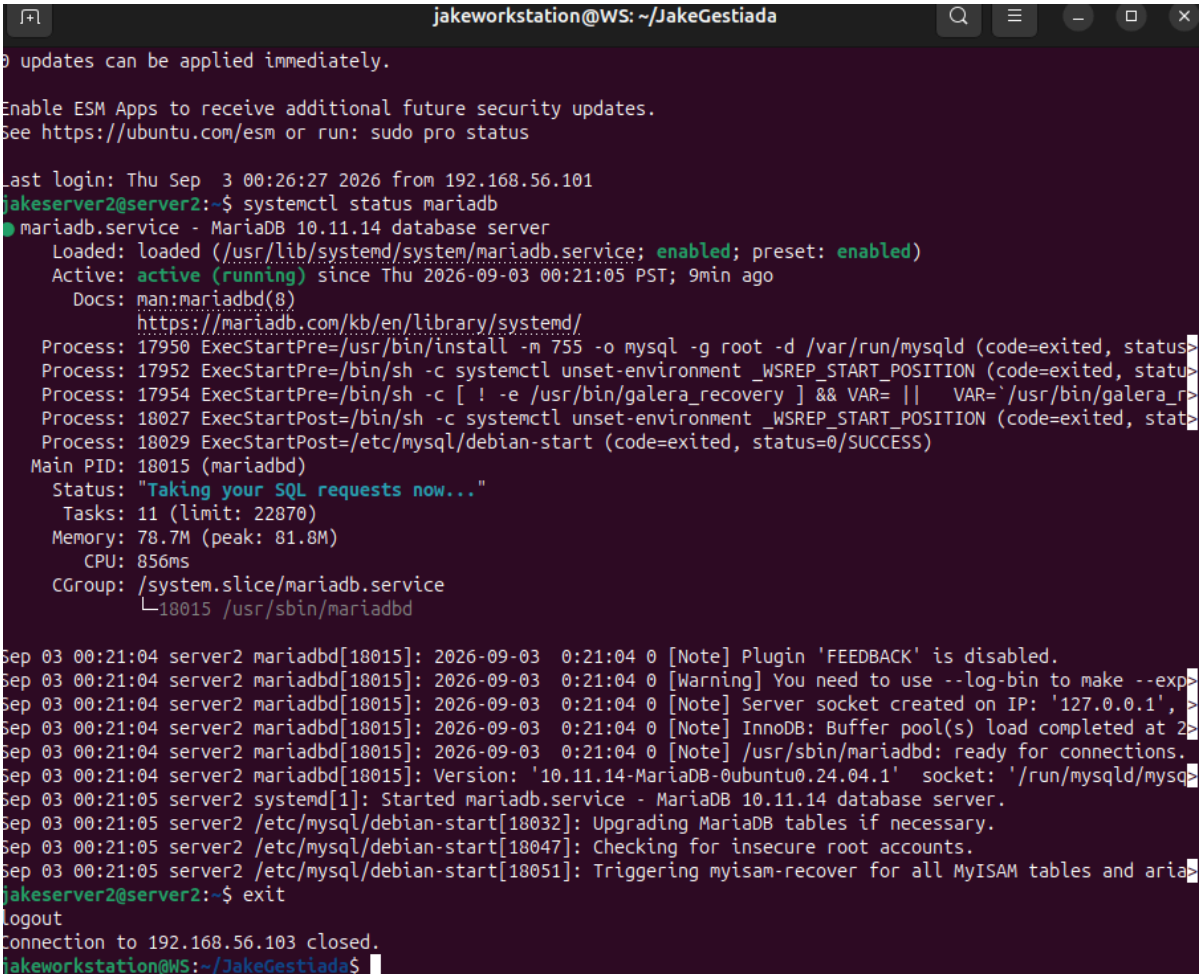


Describe the result:

- Once the playbook was launched, the servers got updated, web servers were set up, and the database servers were ready with a setup involving installation and starting of MariaDB only on machines in db_servers group.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also.
Describe the output

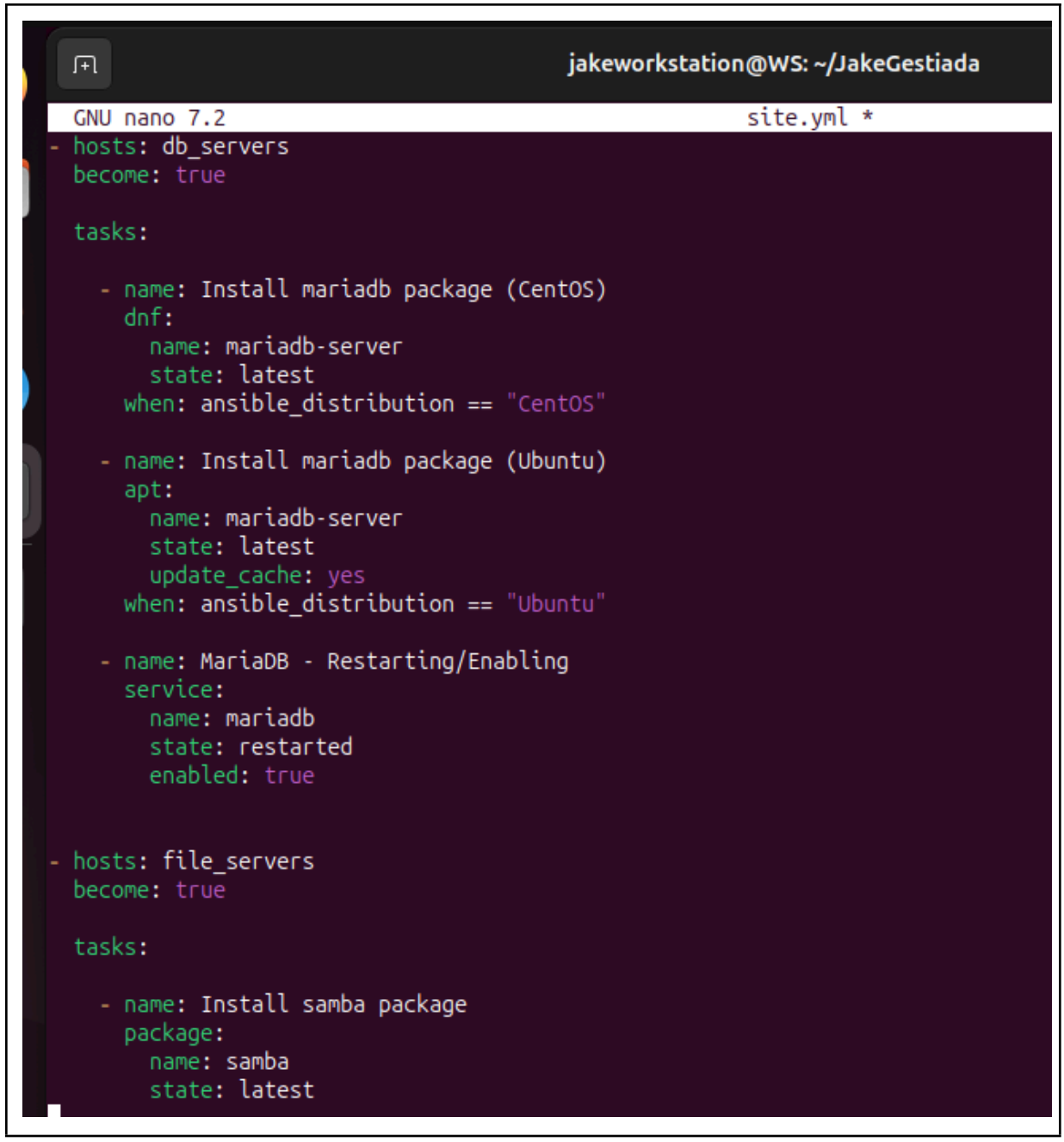

jakeworkstation@WS:~/JakeGestiada$ ssh jakecentos@192.168.56.104
Activate the web console with: systemctl enable --now cockpit.socket
Last login: Thu Sep 3 00:21:03 2026 from 192.168.56.101
[jakecentos@jakecentos ~]$ systemctl status mariadb
mariadb.service - MariaDB 10.5 database server
Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
Active: active (running) since Thu 2026-09-03 00:21:04 PST; 6min ago
Docs: man:mariadbd(8)
https://mariadb.com/kb/en/library/systemd/
Process: 87783 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
Process: 87805 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCE
Process: 87859 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
Main PID: 87840 (mariadbd)
Status: "Taking your SQL requests now..."
Tasks: 8 (limit: 24567)
Memory: 67.5M (peak: 68.5M)
CPU: 373ms
CGroup: /system.slice/mariadb.service
└87840 /usr/libexec/mariadbd --basedir=/usr
Sep 03 00:21:04 jakecentos systemd[1]: Starting MariaDB 10.5 database server...
Sep 03 00:21:04 jakecentos mariadb-prepare-db-dir[87805]: Database MariaDB is probably initialized in /var/lib/m
Sep 03 00:21:04 jakecentos mariadb-prepare-db-dir[87805]: If this is not the case, make sure the /var/lib/mysql
Sep 03 00:21:04 jakecentos systemd[1]: Started MariaDB 10.5 database server.
[jakecentos@jakecentos ~]$
[jakecentos@jakecentos ~]$ ansible all --list-hosts -i inventory.yaml
bash: ansible: command not found...
Install package 'ansible-core' to provide command 'ansible'? [N/y] ^C
[jakecentos@jakecentos ~]$ exit
logout
Connection to 192.168.56.104 closed.
jakeworkstation@WS:~/JakeGestiada$ ansible all --list-hosts -i inventory.yaml

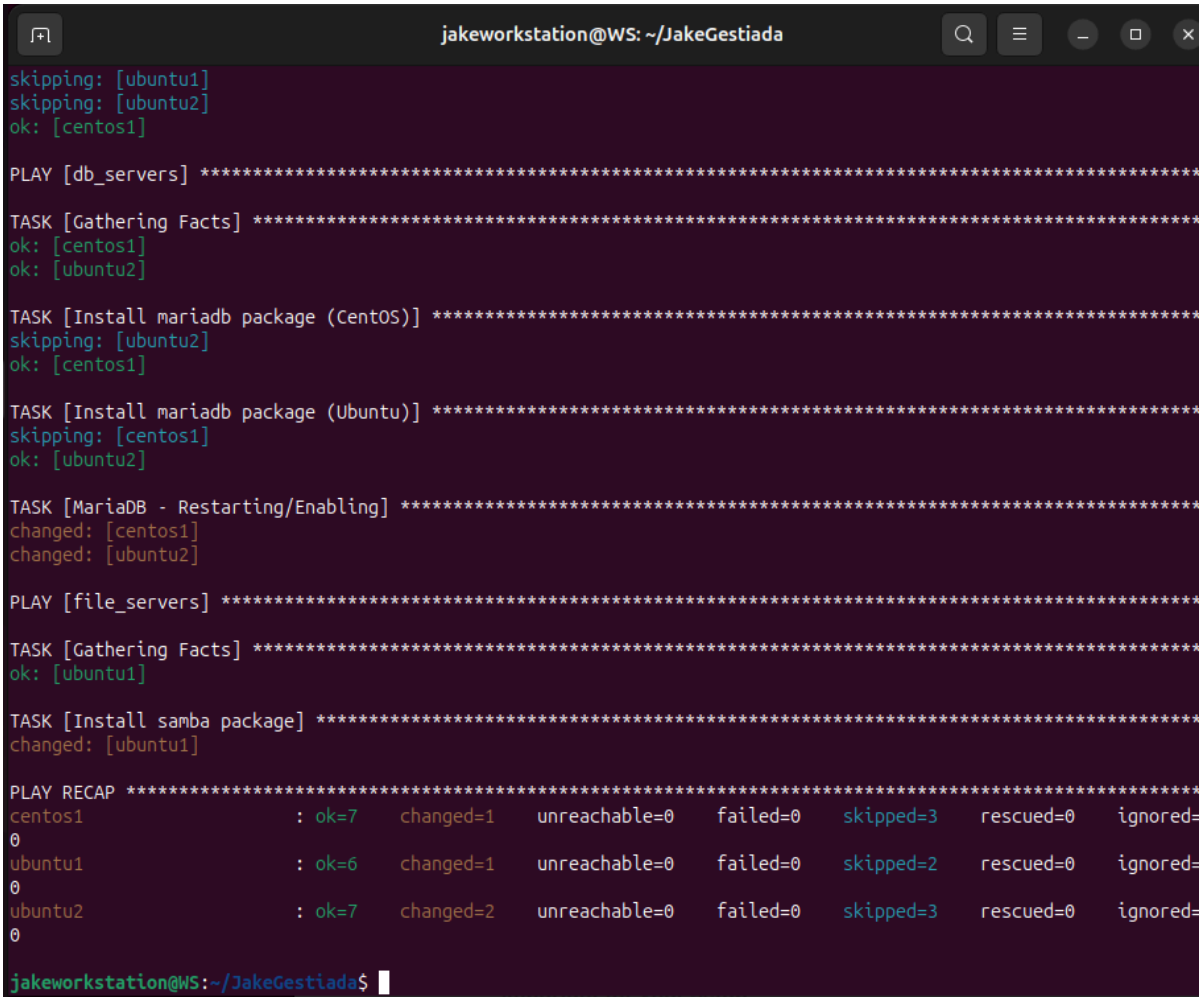


Describe Output:

- MariaDB is present in both the Ubuntu and the CentOS servers. Using the systemctl status command it was confirmed that MariaDB was running fine as a service, i. e., in a started state and that both the database servers had been perfectly configured by Ansible.

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group. We can add the following on our file.
Make sure to save the file and exit.
Run the site.yml file and describe the result.
The testing of the file_servers is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.


jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2 site.yml *
- hosts: db_servers
become: true
tasks:
- name: Install mariadb package (CentOS)
dnf:
name: mariadb-server
state: latest
when: ansible_distribution == "CentOS"
- name: Install mariadb package (Ubuntu)
apt:
name: mariadb-server
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: MariaDB - Restarting/Enabling
service:
name: mariadb
state: restarted
enabled: true
- hosts: file_servers
become: true
tasks:
- name: Install samba package
package:
name: samba
state: latest



Describe the result:

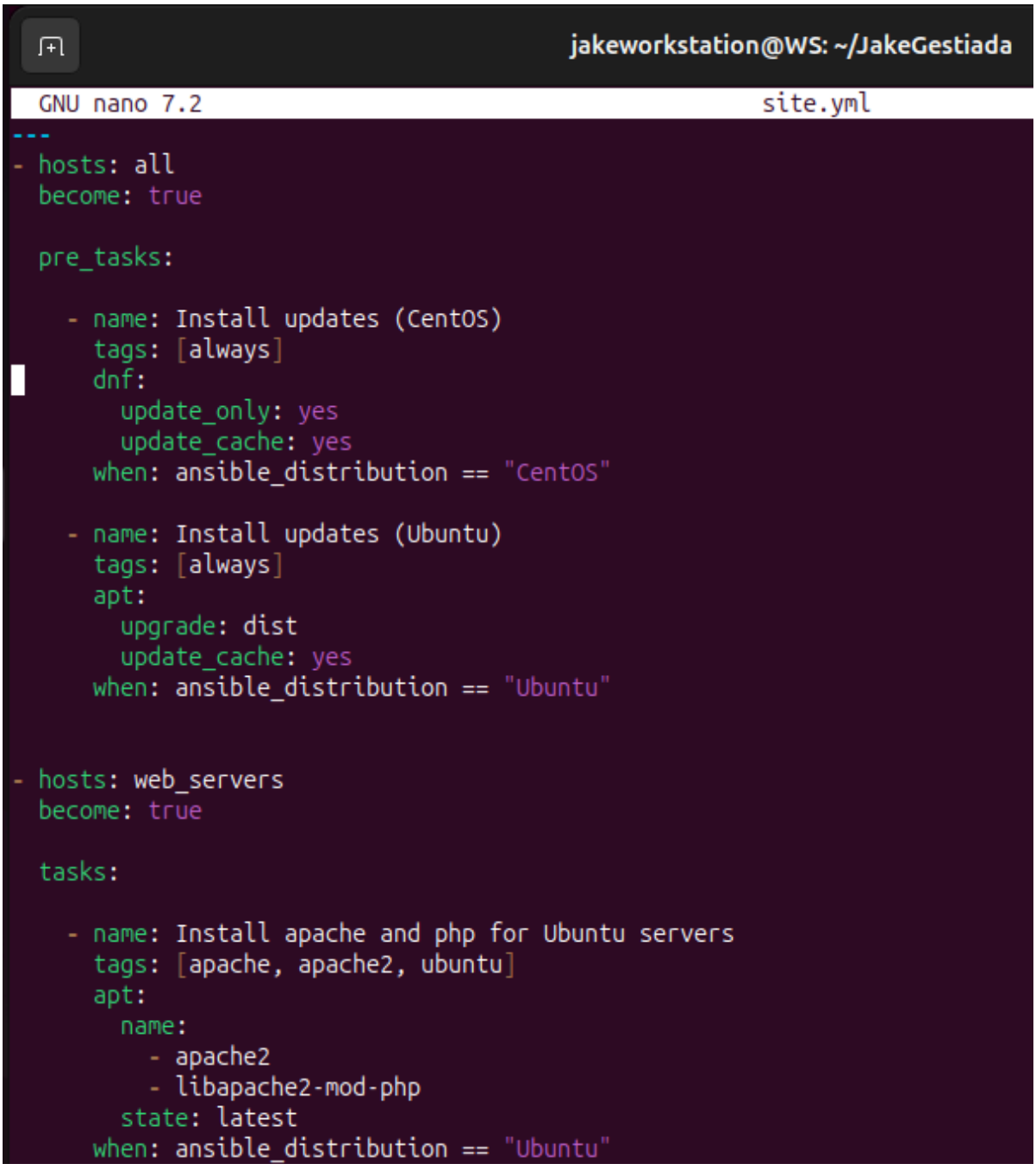
The site.yml playbook worked correctly and selected the file_servers group, and performed the installation task on the server to which it was allocated. This illustrates that Ansible works with groups, and can select specific servers, and apply different configurations to different groups like web_servers, db_servers, and file_servers. There was no need to check the actual file server in this exercise.

Task 2: Using Tags in running playbooks

In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.

1. Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag.

This is an arbitrary command, which means you can use any name for a tag.
Make sure to save the file and exit.
Run the site.yml file and describe the result.




jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2
site.yml
state: latest
when: ansible_distribution == "Ubuntu"
- name: Install apache and php for CentOS servers
tags: [apache, centos, httpd]
dnf:
name:
- httpd
- php
state: latest
when: ansible_distribution == "CentOS"
hosts: db_servers
become: true
tasks:
- name: Install mariadb package (CentOS)
tags: [centos, db, mariadb]
dnf:
name: mariadb-server
state: latest
when: ansible_distribution == "CentOS"
- name: Install mariadb package (Ubuntu)
tags: [db, mariadb, ubuntu]
apt:
name: mariadb-server
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: MariaDB - Restarting/Enabling
tags: [db, mariadb]

```
skipping: [ubuntu2]
skipping: [ubuntu1]
ok: [centos1]

PLAY [db_servers] ***********************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (CentOS)] *************************************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (Ubuntu)] *************************************************************************
skipping: [centos1]
ok: [ubuntu2]

TASK [MariaDB - Restarting/Enabling] ****************************************************************************
changed: [centos1]
changed: [ubuntu2]

PLAY [file_servers] *********************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
ok: [ubuntu1]

TASK [Install samba package] ************************************************************************************
ok: [ubuntu1]

PLAY RECAP ******************************************************************************************************
centos1                    : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored
0
ubuntu1                    : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored
0
ubuntu2                    : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored
0

jakeworkstation@WS:~/JakeGestiada$
```

Describe the result:

- The site.yml playbook finished without any errors and the tags were added to the respective plays. Ansible carried out all three tasks update, web and database server and one additional task file server during the normal execution, without the use of tagging. The main advantage of tags is that we can choose to only run a segment of the playbook if we want it.

2. On the local machine, try to issue the following commands and describe each result:

2.1 ansible-playbook --list-tags site.yml

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

Describe:

The --list-tags option lists all the tags that have been defined in the site.yml playbook without executing the tasks. It presents the existing tags like update, web, db, and fileserver which can subsequently be used for running only the chosen part of the playbook.

2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml --tags centos --ask-become-pass site.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install updates (CentOS)] ************************************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Install updates (Ubuntu)] ************************************************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

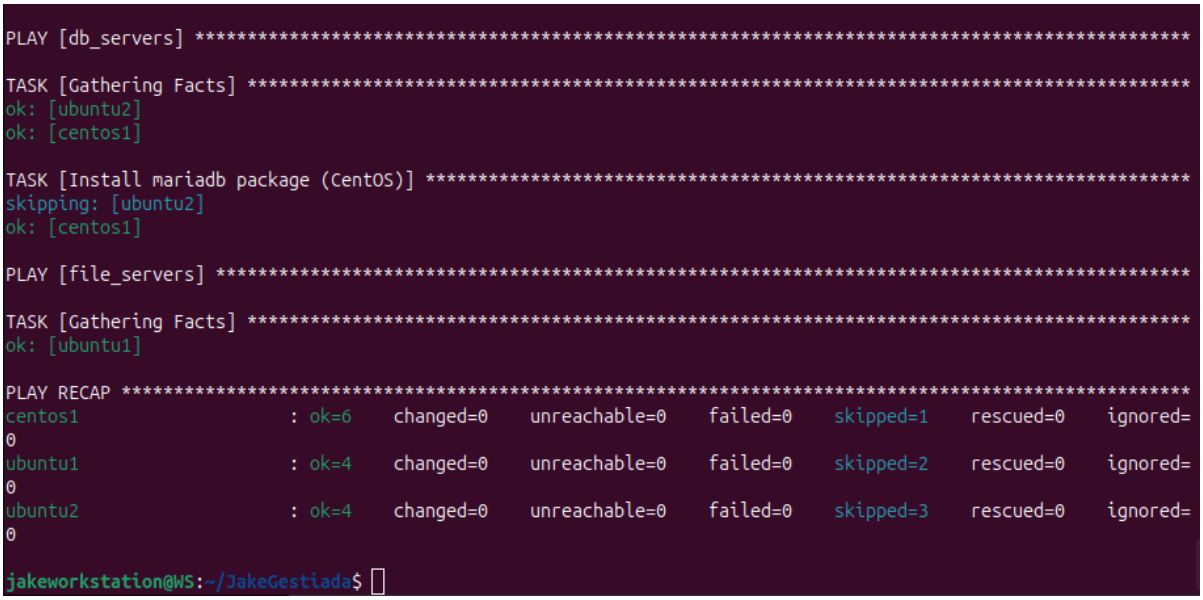
PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache and php for CentOS servers] *******************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [centos1]
```



Describe:

- The command runs only the tasks that have the centos tag. Based on the current site.yml, there are no tasks tagged centos, so Ansible skips the plays and no CentOS configuration tasks are executed.

2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml --tags centos --ask-become-pass site.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install updates (CentOS)] ************************************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Install updates (Ubuntu)] ************************************************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [centos1]
ok: [ubuntu1]

TASK [Install apache and php for CentOS servers] *******************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [centos1]
```

```
PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [centos1]
ok: [ubuntu2]

TASK [Install mariadb package (CentOS)] ****************************************************************************
skipping: [ubuntu2]
ok: [centos1]

PLAY [file_servers] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu1]

PLAY RECAP *********************************************************************************************************
centos1                    : ok=6    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    igno
0
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    igno
0
ubuntu2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    igno
0

jakeworkstation@WS:~/JakeGestiada$ s
```

Describe:

- The command runs only the tasks under the db tag. Ansible targets the db_servers group and

performs the database configuration, including installing MariaDB and starting/enabling the MariaDB service.

2.4 ansible-playbook --tags apache --ask-become-pass site.yml

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml --tags db --ask-become-pass site.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu1]
ok: [centos1]
ok: [ubuntu2]

TASK [Install updates (CentOS)] ************************************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Install updates (Ubuntu)] ************************************************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (CentOS)] ****************************************************************************
skipping: [ubuntu2]
```

```
PLAY [web_servers] ************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

PLAY [db_servers] *************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (CentOS)] ***************************************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (Ubuntu)] ***************************************************************************
skipping: [centos1]
ok: [ubuntu2]

TASK [MariaDB - Restarting/Enabling] ******************************************************************************
changed: [ubuntu2]
changed: [centos1]

PLAY [file_servers] ***********************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [ubuntu1]

PLAY RECAP ********************************************************************************************************
centos1                    : ok=6    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=
0
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=
0
ubuntu2                    : ok=6    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=
0

jakeworkstation@WS:~/JakeGestiada$
```

Describe:

- The command runs only tasks tagged apache. In the current site.yml, the web play is tagged web, not apache, so the Apache installation tasks are not executed.

2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml --tags "apache,db" --ask-become-pass site.
ml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install updates (CentOS)] ************************************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Install updates (Ubuntu)] ************************************************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache and php for Ubuntu servers] *******************************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Install apache and php for CentOS servers] *******************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [db_servers] **************************************************************************************************
```

```
TASK [Install apache and php for CentOS servers] ********************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [db_servers] ***************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [centos1]
ok: [ubuntu2]

TASK [Install mariadb package (CentOS)] *****************************************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (Ubuntu)] *****************************************************************************
skipping: [centos1]
ok: [ubuntu2]

TASK [MariaDB - Restarting/Enabling] ********************************************************************************
changed: [ubuntu2]
changed: [centos1]

PLAY [file_servers] *************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [ubuntu1]

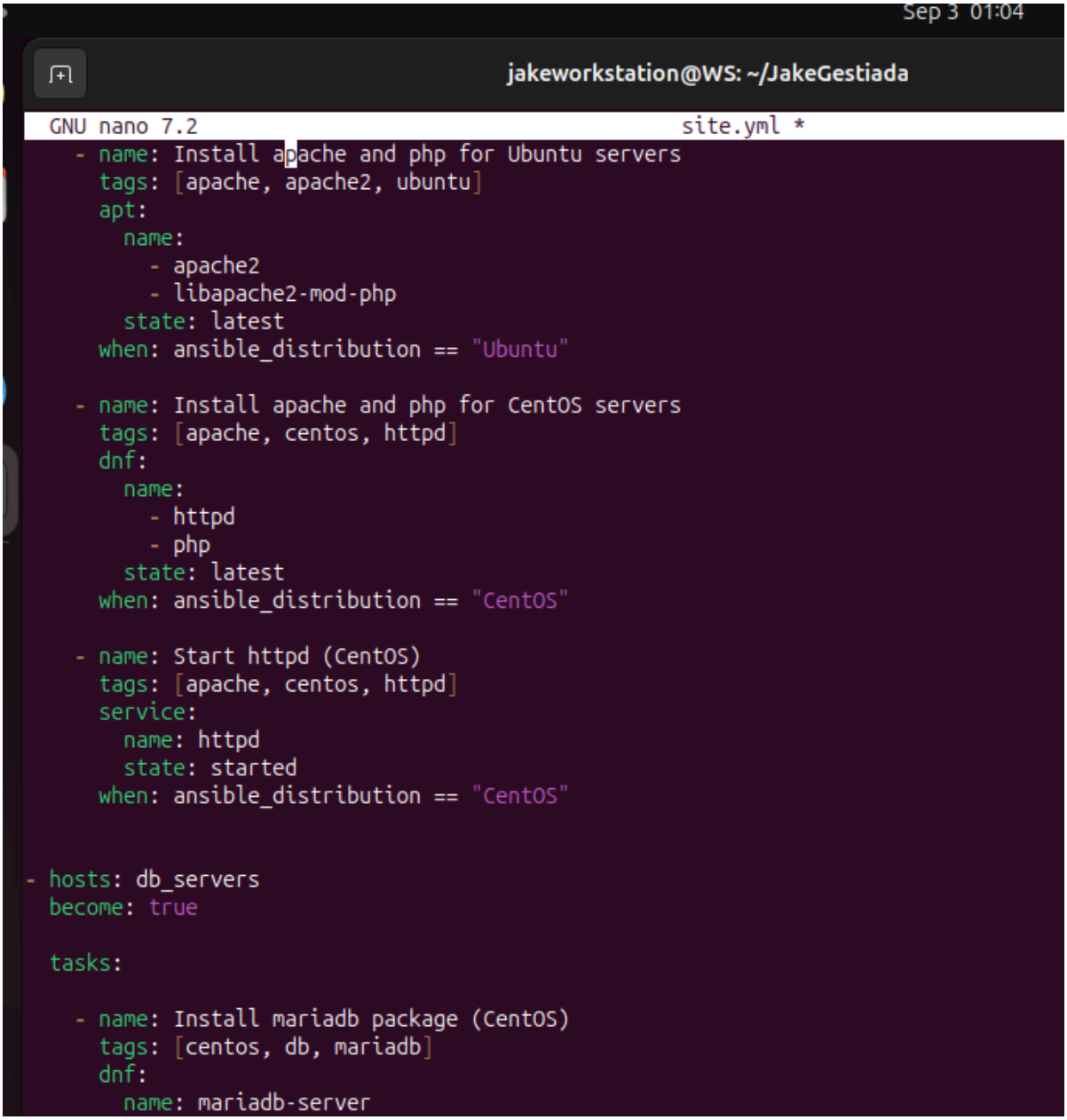
PLAY RECAP **********************************************************************************************************
centos1                    : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=
0
ubuntu1                    : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=
0
ubuntu2                    : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=
0
```

Describe:

- The command tells Ansible to run tasks with either the apache or db tag. In the current site.yml, only the db tag exists, so Ansible runs the database configuration and skips the Apache part because there are no tasks tagged apache.

Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.
Figure 3.1.1
Make sure to save the file and exit.



You would also notice from our previous activity that we already created a module that runs a service.
Figure 3.1.2



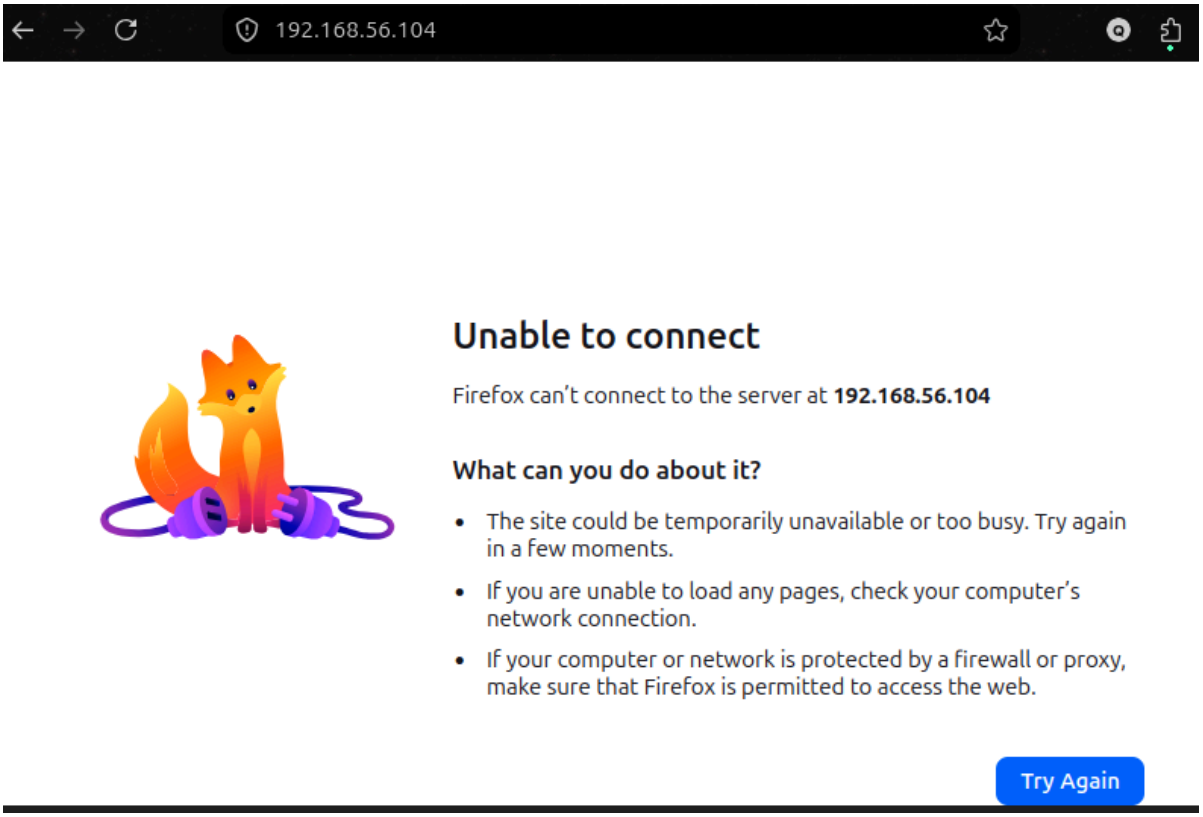
This is because in CentOS, installed packages' services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server's IP address. You should not be getting a display because we stopped the httpd service already.

```
jakeworkstation@WS:~/JakeGestiada$ nano site.yml
jakeworkstation@WS:~/JakeGestiada$ nano site.yml
jakeworkstation@WS:~/JakeGestiada$ ssh jakecentos@192.168.56.104
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Sep  3 00:59:25 2026 from 192.168.56.101
[jakecentos@jakecentos ~]$ sudo systemctl stop httpd
[sudo] password for jakecentos:
[jakecentos@jakecentos ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: inactive (dead) since Thu 2026-09-03 01:07:15 PST; 6s ago
   Duration: 2h 28min 23.257s
       Docs: man:httpd.service(8)
    Process: 976 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
   Main PID: 976 (code=exited, status=0/SUCCESS)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
        CPU: 10.920s

Sep 02 22:38:50 jakecentos systemd[1]: Starting The Apache HTTP Server...
Sep 02 22:38:51 jakecentos httpd[976]: AH00558: httpd: Could not reliably determine the server's fully quali>
Sep 02 22:38:51 jakecentos systemd[1]: Started The Apache HTTP Server.
Sep 02 22:38:51 jakecentos httpd[976]: Server configured, listening on: port 80
Sep 03 01:07:14 jakecentos systemd[1]: Stopping The Apache HTTP Server...
Sep 03 01:07:15 jakecentos systemd[1]: httpd.service: Deactivated successfully.
Sep 03 01:07:15 jakecentos systemd[1]: Stopped The Apache HTTP Server.
Sep 03 01:07:15 jakecentos systemd[1]: httpd.service: Consumed 10.920s CPU time.
lines 1-20/20 (END)
```



3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to

the CentOS server and enter its IP address on the browser. Describe the result.
To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install updates (CentOS)] ************************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Install updates (Ubuntu)] ************************************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

PLAY [web_servers] *************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Install apache and php for Ubuntu servers] *******************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Install apache and php for CentOS servers] *******************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]
```

```
skipping: [ubuntu1]
skipping: [ubuntu2]
changed: [centos1]

PLAY [db_servers] ***********************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
ok: [centos1]
ok: [ubuntu2]

TASK [Install mariadb package (CentOS)] *************************************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [Install mariadb package (Ubuntu)] *************************************************************************
skipping: [centos1]
ok: [ubuntu2]

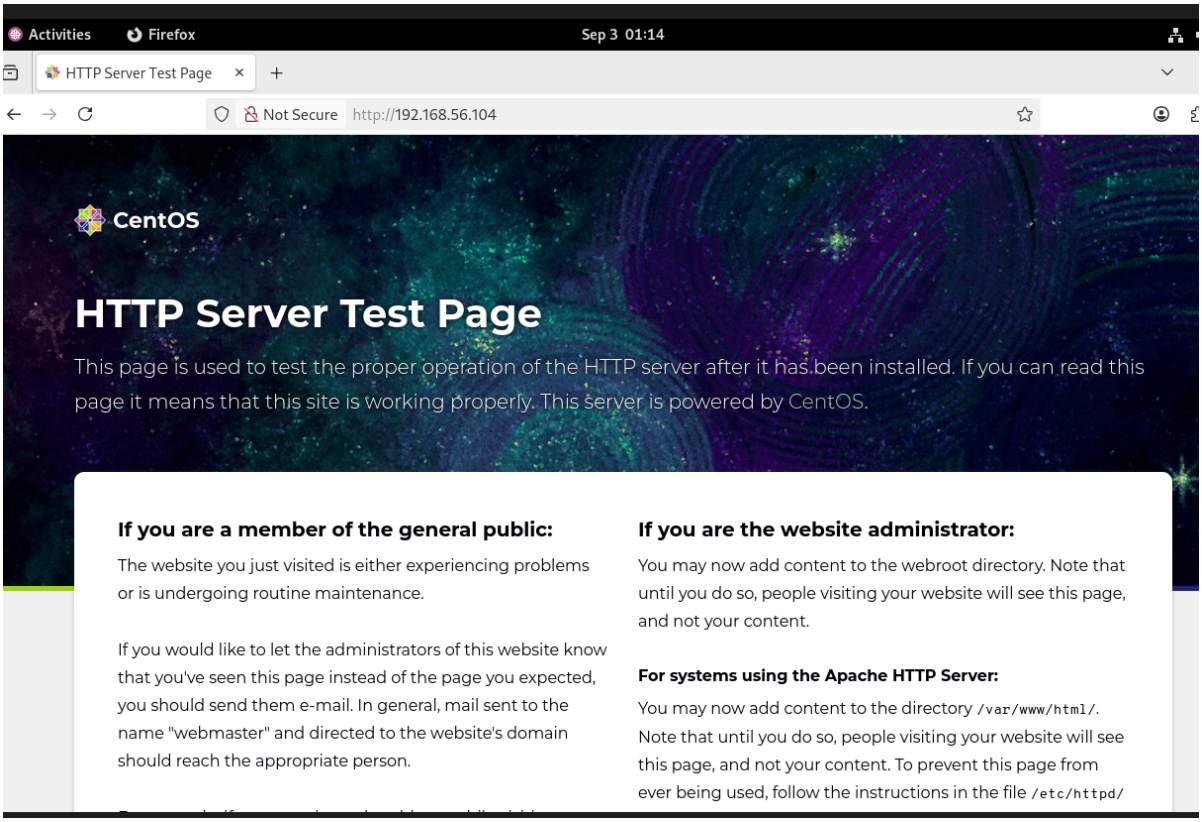
TASK [MariaDB - Restarting/Enabling] ****************************************************************************
changed: [centos1]
changed: [ubuntu2]

PLAY [file_servers] *********************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
ok: [ubuntu1]

TASK [Install samba package] ************************************************************************************
ok: [ubuntu1]

PLAY RECAP ******************************************************************************************************
centos1                    : ok=8    changed=2    unreachable=0    failed=0    skipped=3    rescued=0    igno
red=0
ubuntu1                    : ok=6    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    igno
red=0
ubuntu2                    : ok=7    changed=1    unreachable=0    failed=0    skipped=4    rescued=0    igno
red=0
```



Describe the result:

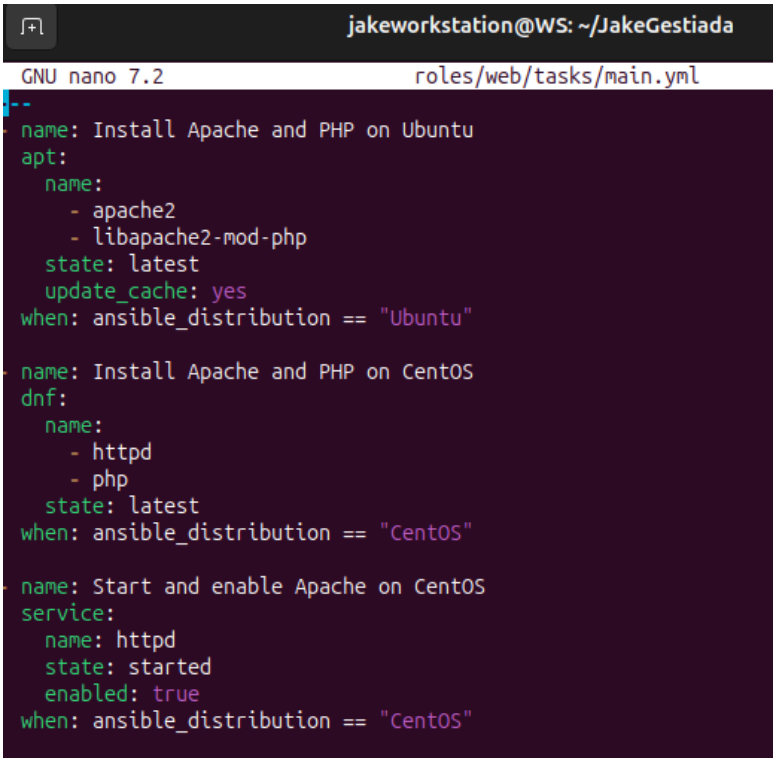
- Once the httpd service was stopped on the CentOS server, the page could not be reached. The site.yml playbook being run caused the Ansible to automatically launch httpd service on the CentOS server. The site became available once again via the CentOS IP address. Besides this, setting the 'enabled: true' value also ensures the service will start automatically upon server boot.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

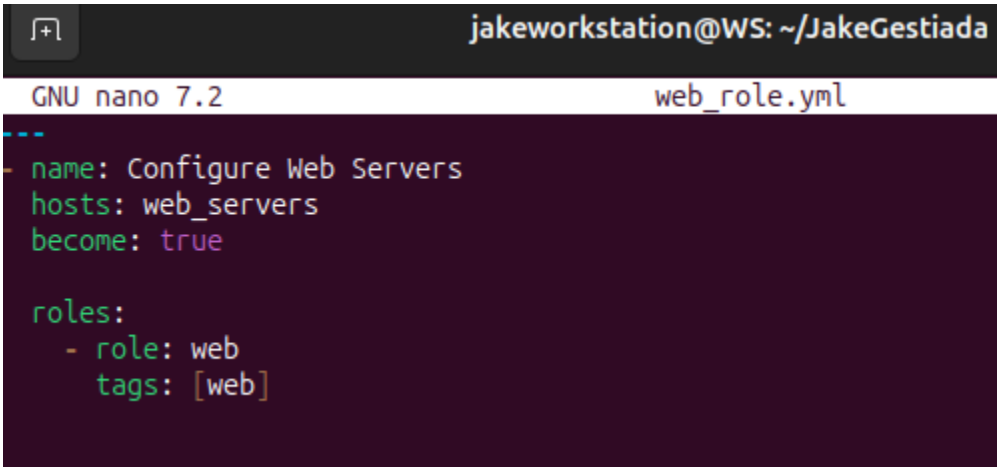
Copy and paste the questions from the lab and include your answers after.

1. Creating a Role for Web Servers

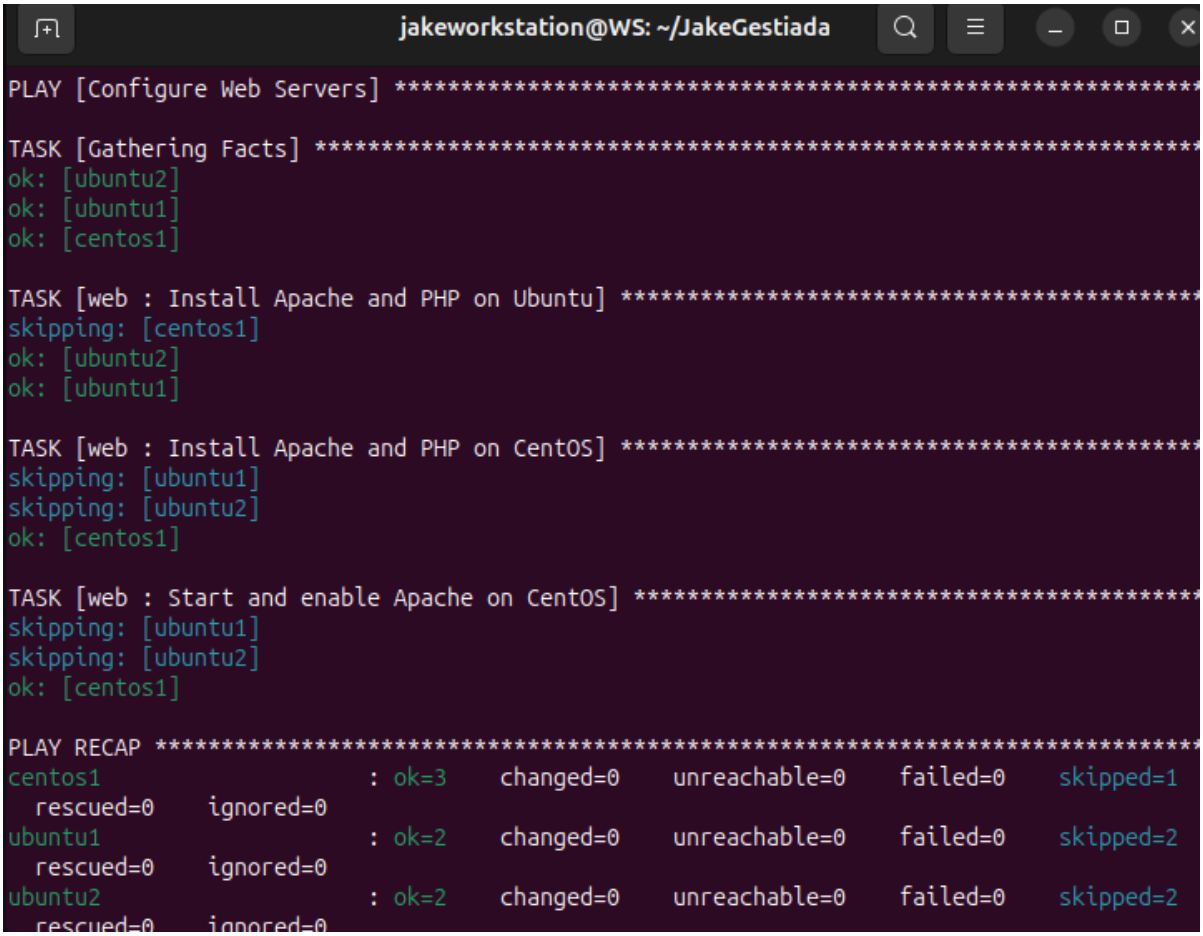
```
jakeworkstation@WS:~/JakeGestiada$ mkdir -p roles/web/tasks
jakeworkstation@WS:~/JakeGestiada$ nano roles/web/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada$ nano web_role.yml
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml web_role.yml --ask-b
ecome-pass
BECOME password:
```


jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2 roles/web/tasks/main.yml

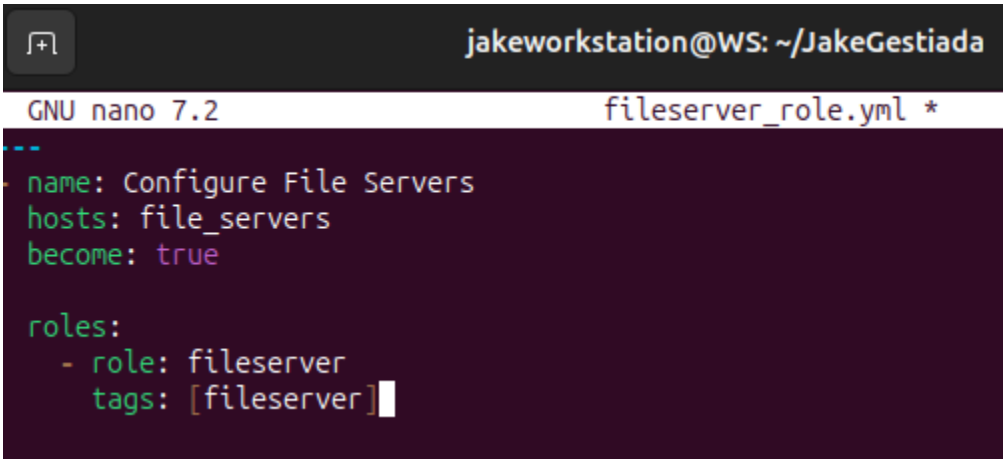
- name: Install Apache and PHP on Ubuntu
apt:
name:
- apache2
- libapache2-mod-php
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: Install Apache and PHP on CentOS
dnf:
name:
- httpd
- php
state: latest
when: ansible_distribution == "CentOS"
- name: Start and enable Apache on CentOS
service:
name: httpd
state: started
enabled: true
when: ansible_distribution == "CentOS"


jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2 web_role.yml

- name: Configure Web Servers
hosts: web_servers
become: true
roles:
- role: web
tags: [web]



2. Creating a Role for file server Servers



```
jakeworkstation@WS:~/JakeGestiada$ nano roles/fileserver/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml fileserver_role.yml
--ask-become-pass
ERROR! the playbook: fileserver_role.yml could not be found
jakeworkstation@WS:~/JakeGestiada$ nano fileserver_role.yml
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml fileserver_role.yml
--ask-become-pass
BECOME password:

PLAY [Configure File Servers] ***************************************************

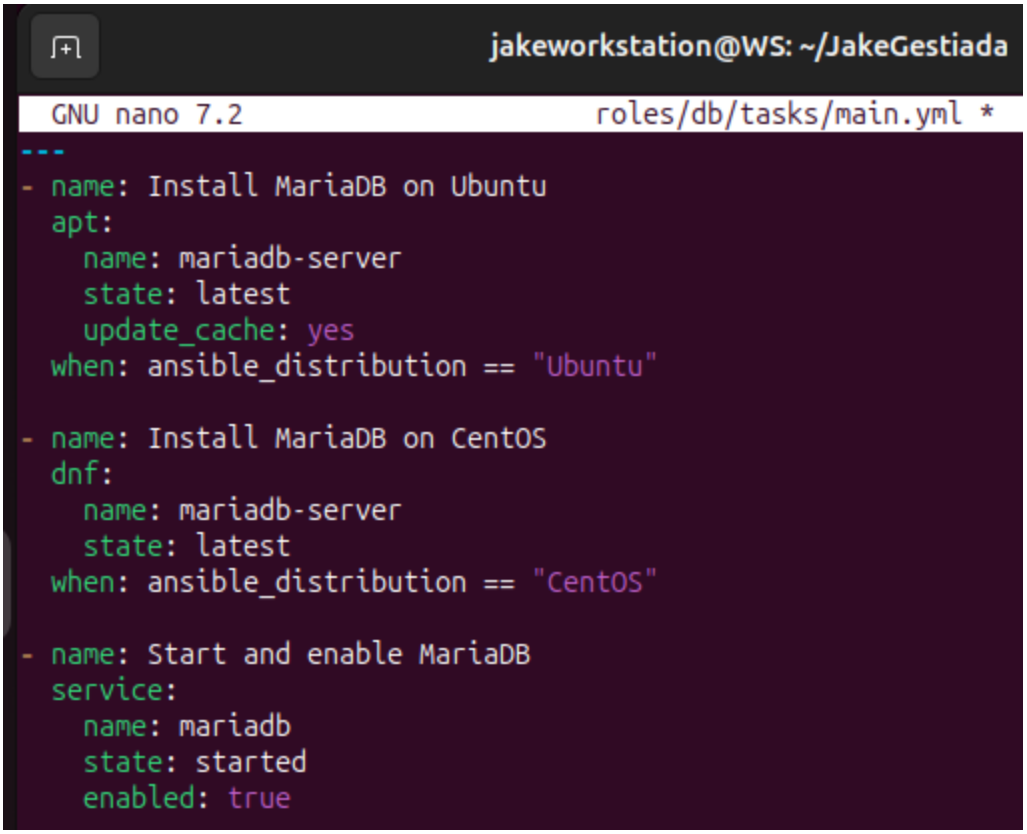
TASK [Gathering Facts] **********************************************************
ok: [ubuntu1]

TASK [fileserver : Install Samba] ***********************************************
ok: [ubuntu1]

PLAY RECAP **********************************************************************
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0
  rescued=0    ignored=0

jakeworkstation@WS:~/JakeGestiada$
```

3. Creating a Role for Database Servers


jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2 roles/db/tasks/main.yml *

- name: Install MariaDB on Ubuntu
apt:
name: mariadb-server
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: Install MariaDB on CentOS
dnf:
name: mariadb-server
state: latest
when: ansible_distribution == "CentOS"
- name: Start and enable MariaDB
service:
name: mariadb
state: started
enabled: true

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml db_role.yml --ask-be
come-pass
BECOME password:

PLAY [Configure Database Servers] ***************************************************

TASK [Gathering Facts] **************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [db : Install MariaDB on Ubuntu] ***********************************************
skipping: [centos1]
ok: [ubuntu2]

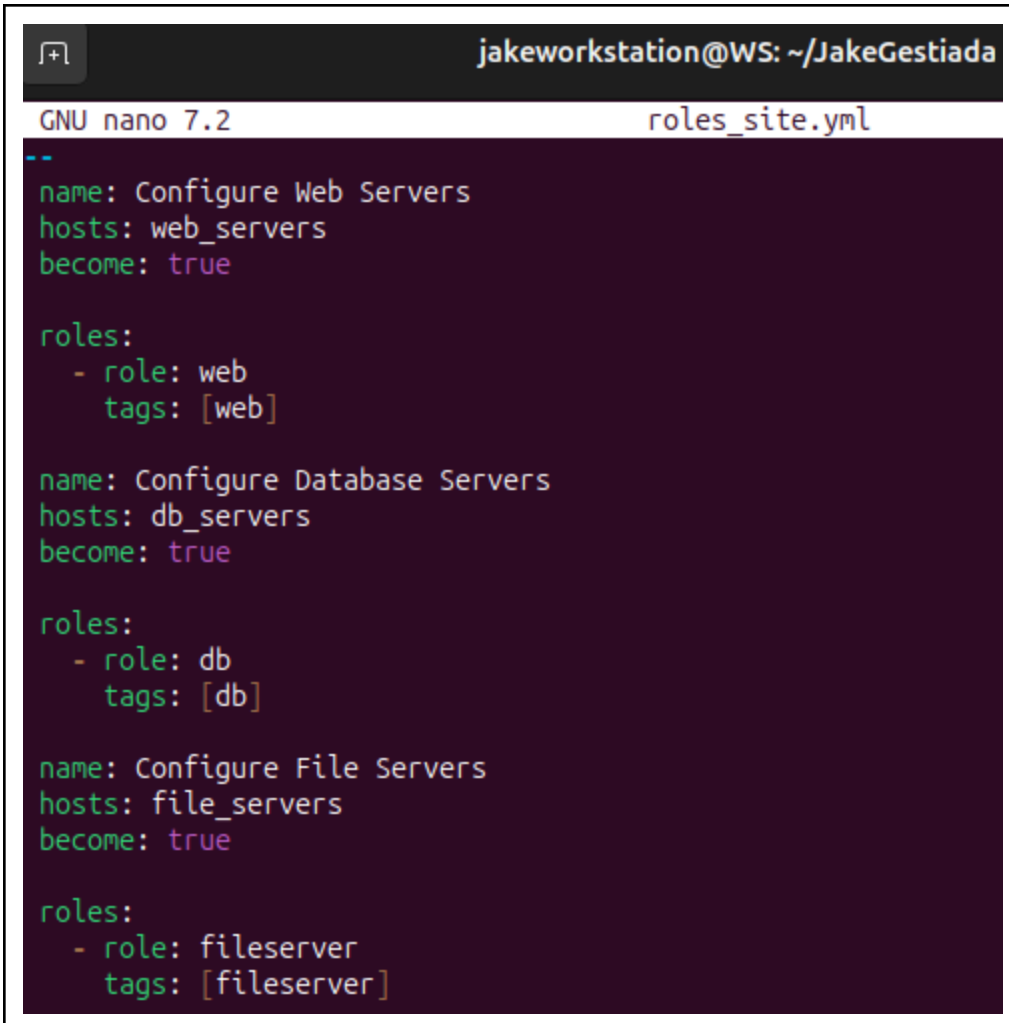
TASK [db : Install MariaDB on CentOS] ***********************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [db : Start and enable MariaDB] ************************************************
ok: [centos1]
ok: [ubuntu2]

PLAY RECAP **************************************************************************
centos1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=1
  rescued=0    ignored=0
ubuntu2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=1
  rescued=0    ignored=0
```

4. Combining Multiple Roles in a Playbook

To orchestrate multiple roles in a single playbook for different server groups. Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for database.


jakeworkstation@WS: ~/JakeGestiada
GNU nano 7.2 roles_site.yml
---
name: Configure Web Servers
hosts: web_servers
become: true
roles:
- role: web
tags: [web]
name: Configure Database Servers
hosts: db_servers
become: true
roles:
- role: db
tags: [db]
name: Configure File Servers
hosts: file_servers
become: true
roles:
- role: fileserver
tags: [fileserver]

```
playbook: roles_site.yml
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml roles_site.yml --ask-b
ome-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu1]
ok: [centos1]
ok: [ubuntu2]

TASK [web : Install Apache and PHP on Ubuntu] **********************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

TASK [web : Install Apache and PHP on CentOS] **********************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [web : Start and enable Apache on CentOS] *********************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]
```

```
PLAY [Configure Database Servers] **********************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [db : Install MariaDB on Ubuntu] ******************************************************
skipping: [centos1]
ok: [ubuntu2]

TASK [db : Install MariaDB on CentOS] ******************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [db : Start and enable MariaDB] *******************************************************
ok: [ubuntu2]
ok: [centos1]

PLAY [Configure File Servers] **************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu1]

TASK [fileserver : Install Samba] **********************************************************
ok: [ubuntu1]
```

```
TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]

TASK [fileserver : Install Samba] **********************************************
ok: [ubuntu1]

PLAY RECAP *********************************************************************
centos1                    : ok=6    changed=0    unreachable=0    failed=0    skipped=2    re
scued=0    ignored=0
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    re
scued=0    ignored=0
ubuntu2                    : ok=5    changed=0    unreachable=0    failed=0    skipped=3    re
scued=0    ignored=0
```

Test web tag:

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml roles_site.yml --tags web
 --ask-become-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos1]

TASK [web : Install Apache and PHP on Ubuntu] **********************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

TASK [web : Install Apache and PHP on CentOS] **********************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [web : Start and enable Apache on CentOS] *********************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [Configure Database Servers] **********************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu2]
ok: [centos1]

PLAY [Configure File Servers] **************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]
```

```
PLAY [Configure File Servers] *************************************************************

TASK [Gathering Facts] ********************************************************************
ok: [ubuntu1]

PLAY RECAP ********************************************************************************
centos1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=1    res
cued=0    ignored=0
ubuntu1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    res
cued=0    ignored=0
ubuntu2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    res
cued=0    ignored=0
```

Test db tag:

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml roles_site.yml --tags db --ask-become-pass
BECOME password:

PLAY [Configure Web Servers] **************************************************************

TASK [Gathering Facts] ********************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

PLAY [Configure Database Servers] *********************************************************

TASK [Gathering Facts] ********************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [db : Install MariaDB on Ubuntu] *****************************************************
skipping: [centos1]
ok: [ubuntu2]

TASK [db : Install MariaDB on CentOS] *****************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [db : Start and enable MariaDB] ******************************************************
ok: [ubuntu2]
ok: [centos1]

PLAY [Configure File Servers] *************************************************************

TASK [Gathering Facts] ********************************************************************
ok: [ubuntu1]

PLAY RECAP ********************************************************************************
centos1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored
=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored
=0
```

```
TASK [Gathering Facts] ********************************************************************
ok: [ubuntu1]

PLAY RECAP ********************************************************************************
centos1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

Test fileserver tag:

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml roles_site.yml --tags fileserver --ask-become
-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos1]

PLAY [Configure Database Servers] **********************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [centos1]

PLAY [Configure File Servers] **************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu1]

TASK [fileserver : Install Samba] **********************************************************************************
ok: [ubuntu1]

PLAY RECAP *********************************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

List the available list

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml roles_site.yml --list-tags

playbook: roles_site.yml

  play #1 (web_servers): Configure Web Servers  TAGS: []
      TASK TAGS: [web]

  play #2 (db_servers): Configure Database Servers      TAGS: []
      TASK TAGS: [db]

  play #3 (file_servers): Configure File Servers        TAGS: []
      TASK TAGS: [fileserver]
```

Test multiple tags together:

```
jakeworkstation@WS:~/JakeGestiada$ ansible-playbook -i inventory.yaml roles_site.yml --tags "web,db" --ask-become-p
ass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos1]

TASK [web : Install Apache and PHP on Ubuntu] **********************************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

TASK [web : Install Apache and PHP on CentOS] **********************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [web : Start and enable Apache on CentOS] *********************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY [Configure Database Servers] **********************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [db : Install MariaDB on Ubuntu] ******************************************************************************
skipping: [centos1]
ok: [ubuntu2]

TASK [db : Install MariaDB on CentOS] ******************************************************************************
skipping: [ubuntu2]
```

```
TASK [db : Install MariaDB on CentOS] ******************************************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [db : Start and enable MariaDB] *******************************************************************************
ok: [ubuntu2]
ok: [centos1]

PLAY [Configure File Servers] **************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [ubuntu1]

PLAY RECAP *********************************************************************************************************
centos1                    : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

ubuntu1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

ubuntu2                    : ok=5    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

Reflections:

Answer the following:

1. What is the importance of putting our remote servers into groups?

- Server management is more convenient, and task allocation goes hand in hand. Take, for one, that web-related works are assigned to the web-serving machines while database chores are given to the ones with database servers.

2. What is the importance of tags in playbooks?

- Tags make it possible for us to select and run only the specific tasks. That is time efficient because we don't have to run the whole playbook just for a single part.

3. Why do think some services need to be managed automatically in playbooks?

- Some services don't start automatically after an installation and the only way is to start and enable them manually. By including them in the playbook the services are going to automatically run and stay on without any manual intervention from our side.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **This helps to identify the errors that can't run.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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